

Which Captions Should Be Positive for Image–Text Retrieval?

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Abstract

An image can have several correct descriptions. Which additional captions should a model learn to match? We compare supported descriptions with random choices of the same number of training targets. All candidate captions stay fixed across 72 training runs of one pretrained model with small learned corrections. We test top-ranked matches among 1,000 images and 5,000 original captions. Compared with random choices within initial-similarity groups, supported labels improve finding the correct image from a caption by 1.54 to 4.08 percentage points. Compared with training to match only original captions, finding those captions from an image falls by 6.00 to 10.83 points. Supported labels also improve tests with edited wrong captions and valid rewordings over that baseline. A retrieval-based development strategy selects the unadapted model for every policy that adds training targets. These controlled comparisons separate caption identity, target count, training schedule, and model selection.

1 Introduction

A photo of a child holding a red ball supports several descriptions. “A child with a red ball” describes a specific scene. “Someone is outdoors” may also be correct, but apply to many images. An image search model must learn from such valid descriptions while still finding the intended match among thousands of candidates. This raises a practical question: which additional descriptions should training reward?

Contrastive image–text models assign a score to each image and caption. Training marks associated captions as *positives* and rewards their scores relative to competing candidates (Radford et al., 2021). A competing caption need not be false. It may be another valid description of the image. Relation annotations distinguish statements supported by an image from contradicted or unresolved statements.

Promoting the supported statements to positives is a natural use of those labels.

That intervention changes more than correctness. It increases the number of positive targets per image and the number of captions trained to retrieve an image. It can also change which learning rate or epoch performs best on development data. To determine what the supported labels contribute, we need comparisons that separate caption identity from these changes.

Prior work shows that better rejection of edited wrong captions can coexist with worse handling of valid rewordings (Kamath et al., 2024). FSC-CLIP studies retention of broader capabilities, while CLIC and concept-centric learning improve compositional and retained performance together (Oh et al., 2024; Peleg et al., 2025; Pham et al., 2026). Related methods convert false negatives into positives, regulate negative weights, or divide candidates into positive, negative, and neutral groups (Byun et al., 2024; Jiang et al., 2023; Liu et al., 2023). Our question concerns the assignment of *additional positive captions* within an unchanged candidate pool.

We study one frozen image–text encoder with small trainable corrections. Supported promotion is compared with source-positive training and two randomized promotion controls. The controls preserve positive counts; one also preserves counts within coarse initial-similarity groups. We compare every learning rate at the same epoch, then separately evaluate two development-selection strategies. Both retrieval directions use the complete, disjoint source-caption test pool. This design connects the value of the annotation to the actual search task.

2 A Controlled Positive-Assignment Study

The candidate captions never change. Each training image has five original source captions and hypotheses from e-SNLI-VE (Kayser et al., 2021). The hypotheses carry supported, contradicted, or neutral labels. A batch contains its images and all their source captions and hypotheses. Every policy scores exactly these candidates. A caption that is not positive remains a competitor, including supported hypotheses under the source baseline.

Source makes only the five source captions positive. *Supported* additionally promotes every supported hypothesis for its annotated image. *Count-only* instead selects the same number of that image’s hypotheses uniformly at random. *Score-stratified* ranks the image’s hypotheses by frozen image–text similarity, divides them into two nearly equal groups, and randomly selects the supported count within each group. It matches coarse initial difficulty as well as count. The controls use three independent assignments each, fixed throughout fitting. These are label-informed controls because their counts come from the supported annotations.

All promoted captions have one owning image. Thus the three expanded policies have the same positive count for each image and the same number of captions that serve as text-to-image training queries. They differ in *which* hypotheses are positive. No contradiction penalty or neutral exclusion is applied in this study. The precise assignment rule and its invariants appear in Appendix A.2.

Matched model and training schedules. The encoder is OpenCLIP ViT-B/32, pretrained on LAION-2B (Cherti et al., 2023). Its image and text features stay fixed. Two zero-initialized linear corrections, one for each feature type, learn 524,288 parameters. Corrected features are normalized and scored by scaled cosine similarity. The symmetric contrastive objective gives equal target probability to each positive, averaging image-query and eligible caption-query losses (Khosla et al., 2020).

The same 1,200 training images, image order, optimizer, and ten-epoch schedule are used for every condition. Two deterministic policies and six randomized assignments are crossed with three learning rates (10^{-4} , 3×10^{-4} , 10^{-3}) and three training seeds. This yields 72 fits. Our primary comparison uses epoch ten at each rate; it never chooses a rate using test performance. Early epochs and all individual runs are retained. Appendix A.4 specifies

the optimizer and exact replay checks against the prior source and supported fits.

Search the complete source pool. For an image query, the task is to find one of its five original captions: image-to-text retrieval (I→T). For a caption query, the task is to find its owning image: text-to-image retrieval (T→I). Recall at rank one (R@1) is the percentage of queries whose highest-scoring candidate is a relevant match.

We use all 1,000 official e-ViL test images and their 5,000 source captions, with no training, calibration, or development image overlap. Both directions search the full pool. The hypotheses used for relation evaluation do not enter this retrieval gallery. This source-domain test adds 600 images to the 400-image relation-test subset retained from the prior study. Appendix A.5 records the split and byte-level overlap audits.

Inference and complementary tasks. The primary family contains all supported-minus-comparator contrasts at three rates in both retrieval directions: 18 effects. Random controls average three assignment draws within each training seed, then the three paired seed differences. We use 10,000 paired image-cluster resamples with an eighteen-comparison adjustment. Figure 1 shows every effect and its conditional 99.72% interval; all five caption queries stay with their image cluster.

Descriptive tests assess what the additional labels teach beyond source retrieval. Relation accuracy measures supported-versus-contradicted ranking on the original 400 images. SugarCrep tests a correct caption against an edited wrong caption, called a *foil* (Hsieh et al., 2023). SugarCrep++ requires two valid captions to *both* score above the same foil (Dumpala et al., 2024). We also retain full-pool COCO retrieval. All endpoints, trajectories, and draw-level results appear in Appendix B.

3 What Caption Identity Contributes

A text-to-image benefit beyond random promotion. Supported labels outperform both randomized controls on T→I retrieval at every learning rate. All six adjusted intervals lie above zero. The advantage holds in each assignment-draw and training-seed comparison, not just after averaging. The score-stratified control is the stronger attribution comparison: it keeps both the number of promoted captions and their coarse frozen-score distribution. Thus the supported-label assignment

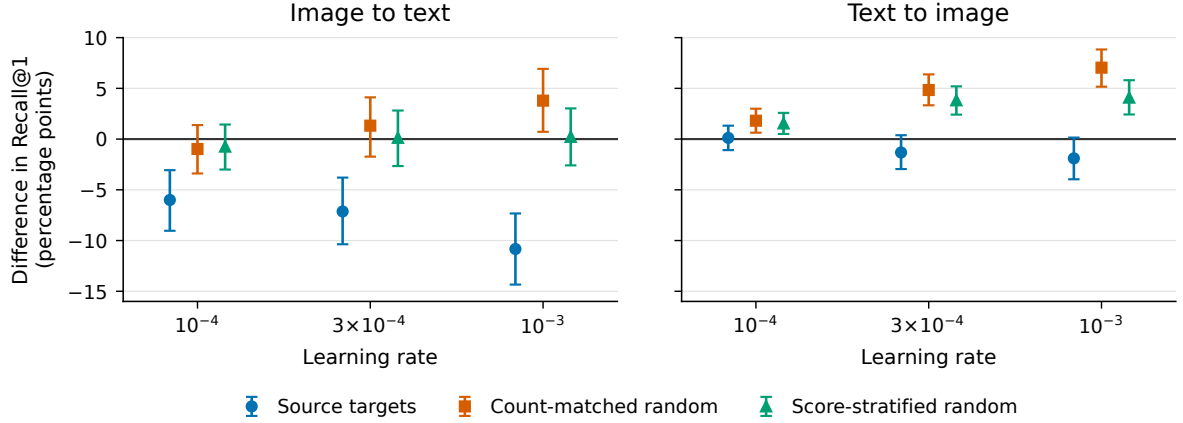


Figure 1: All 18 primary comparisons at matched epoch ten. Points are supported promotion minus the named comparator. Bars are 99.7222% paired image-cluster intervals, adjusted for the 18 comparisons and conditional on the fitted seeds and assignment draws. Random controls average three fixed assignments within each of three training seeds.

helps source captions retrieve their owning images beyond these two matching constraints.

The I→T comparison has a different evidence pattern. Against score-stratified promotion, the three effects are -0.74, +0.12, and +0.22 points, and their intervals cross zero. Against count-only promotion, a positive effect is detected only at the largest learning rate. The observed supported-label advantage is clearest for using captions to find images. Figure 1 keeps the endpoint and comparator visible in every comparison.

The source-caption cost persists under matched schedules. Supported promotion reduces I→T recall relative to Source at all three learning rates; every adjusted interval lies below zero. Averaged over training seeds, this means 60 to 108 extra failed image queries in the 1,000-image pool. Each failure returns a top caption outside the image’s five source captions. The difference is present within the source domain and with identical epoch and learning rate. It therefore persists after removing the schedule and cross-dataset changes of an independently selected comparison. The T→I supported-minus-source intervals cross zero at all three rates.

Better caption discrimination accompanies the retrieval cost. Table 1 reports the matched source and supported models alongside frozen initialization. Supported promotion improves relation accuracy, edited-caption discrimination, and the two-valid-caption endpoint over Source at every rate. The SugarCrepe gains grow from 2.94 to 4.93

Rate	Positives	Relation	Edited	Both
–	Frozen	79.47	81.21	68.95
10^{-4}	Source	77.75	81.00	64.21
10^{-4}	Supported	87.62	83.93	67.75
$3 \cdot 10^{-4}$	Source	75.35	80.14	63.77
$3 \cdot 10^{-4}$	Supported	86.68	83.35	67.16
10^{-3}	Source	70.51	76.35	60.25
10^{-3}	Supported	85.78	81.28	65.62

Table 1: Mean accuracy (%) at matched epoch ten. Relation is e-SNLI-VE; Edited is SugarCrepe; Both requires both valid SugarCrepe++ captions above the foil. These are descriptive endpoints.

points. Both valid captions also fare better than under source-positive adaptation. The frozen row supplies the absolute reference for these matched gains. These tests reward rejecting a specified foil; full retrieval requires placing a source match above every gallery competitor.

The retrieval pattern extends to COCO. The separate full COCO pool contains 5,000 images and 25,000 captions. At the three matched learning rates, supported-minus-source I→T differences are -4.73, -6.12, and -7.21 points. The corresponding T→I differences are -0.53, -0.90, and -0.08 points. These descriptive outcomes extend the source-pool image-query cost to a second image collection. Reporting every rate keeps this comparison aligned with the primary schedule. Appendix B also gives all declared early-epoch results, including both valid captions separately.

4 Selection Can Reject the Learned Expansion

The matched terminal comparison describes the intervention at a fixed schedule. A deployed model also depends on which state development data selects. We evaluate two complete selection strategies over the same saved states, including epoch zero.

The *native* strategy averages relation accuracy with image-to-text retrieval on the prior 100-image validation pool, where either a source or supported caption counts as relevant. The *source-retrieval* strategy averages both retrieval directions on 900 development images and 4,500 original captions, excluding the calibration images. For each condition, learning rate, and training seed, a strategy selects an epoch; it then selects a shared learning rate across seeds. Random assignment draws are selected separately and receive equal reporting weight.

Table 2 shows the source and supported outcomes. Native selection retains trained expansion states. Source-retrieval selection chooses epoch zero for Supported and for all six randomized assignments: all 21 expanded selections return frozen initialization. It retains nonzero epochs for Source. The retrieval-oriented procedure therefore does not produce a trained expansion model that keeps its discrimination gains. It chooses not to use those learned corrections.

These are complete strategies with different relevance definitions, development pools, and direction balances. Their selected models answer a different question from the matched epoch-ten experiment. Reporting both prevents a selected procedure from standing in for the effect of positive assignment itself.

Selection	Positives	Epochs	I→T	T→I
Native	Source	1/1/3	84.70	69.35
Native	Supported	5/10/5	77.00	63.30
Retrieval	Source	1/2/2	84.83	69.70
Retrieval	Supported	0/0/0	84.50	69.76

Table 2: Selected procedures on the full source-caption test pool. Epochs follow seeds 17/29/43. Native supported uses learning rate $3 \cdot 10^{-4}$; the other rows use 10^{-4} . Epoch zero is frozen initialization.

5 Why the Counts Must Be Matched

The controls preserve two quantities because the training objective averages over two different kinds of queries. To make this precise, let s_{ij} be the score of image i and caption j . Let P_i be the nonempty set of its positive captions. With unit candidate weights, the image-query loss is

$$\ell_i = \log \sum_j e^{s_{ij}} - \frac{1}{|P_i|} \sum_{j \in P_i} s_{ij}. \quad (1)$$

The sum inside the logarithm contains every batch caption. The model probability for caption j is $p_{ij} = e^{s_{ij}} / \sum_k e^{s_{ik}}$. Differentiating gives $p_{ij} - 1/|P_i|$ on a positive caption. The target is shared equally, rather than added independently for each positive.

Suppose an image has K source positives and M hypotheses are promoted. Holding scores and candidates fixed, each source target changes from $1/K$ to $1/(K+M)$. With five source captions and five promoted hypotheses, it falls from 0.2 to 0.1. A caption can remain positive while receiving less target probability. This change is identical for Supported, Count-only, and Score-stratified because they promote the same number for that image.

The reverse objective averages over captions with a positive image. Each existing source caption keeps its target on its owning image, but receives a smaller share of the enlarged caption average. Equal promotion counts also match this average’s size across the expanded policies. At fixed scores, their full symmetric score gradients on every original source-caption column are therefore identical. Appendix A.3 gives the exact identity and proof.

The policies differ on hypothesis columns, whose gradients also update the shared image and text corrections. The controls isolate that assignment difference while matching the source-column gradients at common scores. This is the precise role of the count constraints. The observed trained retrieval effects are the empirical part of the comparison.

Implication for retrieval studies. Supported descriptions and retrieval relevance serve different purposes. One states that an image supports a caption; the other asks whether a candidate is an annotated match in a large gallery. Positive-assignment comparisons should name their reference policy and report the two retrieval directions separately. A selected model adds another decision, which the matched-schedule result cannot stand in for.

Limitations

This study uses one frozen OpenCLIP checkpoint, linear residual adaptation, and 1,200 training images. It does not establish behavior under full-encoder training, other objectives, larger data scales, or other backbones. The supported annotations partly inherit text-inference judgments and are not independent visual adjudications of every statement. Source-caption ownership supplies incomplete retrieval relevance. Coarse score strata do not remove every semantic or difficulty difference between assignments.

Primary intervals condition on the three fitted seeds, three assignment draws, and fixed gallery. They resample query-image clusters, not new fitted models or galleries. They do not establish equivalence when spanning zero. The secondary tasks, selected-strategy comparisons, trajectories, and original 400-image and additional 600-image subgroups are descriptive. COCO-derived benchmarks overlap. The two development selectors change several choices jointly, so their contrast does not isolate the score formula alone. Fixed-score algebra describes the objective and matched controls; it is not a causal explanation of learned retrieval losses. The separate allocation experiment described in Appendix A.8 was not run.

Ethical Considerations and Assistance

We use existing public datasets under their source terms and collect no new human judgments. Captions and pretrained features may reflect source biases. Generative AI assisted planning, implementation, analysis, literature discovery, and drafting. The authors are responsible for checking the released computations and manuscript.

References

Jaeseok Byun, Dohoon Kim, and Taesup Moon. 2024. [MAFA: Managing false negatives for vision-language pre-training](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27314–27324.

Mehdi Cherti, Romain Beaumont, Ross Wightman, Mitchell Wortsman, Gabriel Ilharco, Cade Gordon, Christoph Schuhmann, Ludwig Schmidt, and Jenia Jitsev. 2023. [Reproducible scaling laws for contrastive language-image learning](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2818–2829.

Sri Harsha Dumpala, Aman Jaiswal, Chandramouli Sastri, Evangelos Milios, Sageev Oore, and Hassan Sajjad. 2024. [SUGARCREPE++ dataset: Vision-language model sensitivity to semantic and lexical alterations](#). *Preprint*, arXiv:2406.11171.

Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Anirudha Kembhavi, and Ranjay Krishna. 2023. [SugarCrepe: Fixing hackable benchmarks for vision-language compositionality](#). In *Advances in Neural Information Processing Systems*, volume 36.

Chaoya Jiang, Wei Ye, Haiyang Xu, Songfang Huang, Fei Huang, and Shikun Zhang. 2023. [Vision language pre-training by contrastive learning with cross-modal similarity regulation](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14660–14679. Association for Computational Linguistics.

Amita Kamath, Cheng-Yu Hsieh, Kai-Wei Chang, and Ranjay Krishna. 2024. [The hard positive truth about vision-language compositionality](#). In *European Conference on Computer Vision*.

Maxime Kayser, Oana-Maria Camburu, Leonard Salewski, Cornelius Emde, Virginie Do, Zeynep Akata, and Thomas Lukasiewicz. 2021. [e-ViL: A dataset and benchmark for natural language explanations in vision-language tasks](#). In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. 2020. [Supervised contrastive learning](#). In *Advances in Neural Information Processing Systems*, volume 33.

Bo Liu, Donghuan Lu, Dong Wei, Xian Wu, Yan Wang, Yu Zhang, and Yefeng Zheng. 2023. [Improving medical vision-language contrastive pretraining with semantics-aware triage](#). *IEEE Transactions on Medical Imaging*, 42(12):3579–3589.

Youngtaek Oh, Jae Won Cho, Dong-Jin Kim, In So Kweon, and Junmo Kim. 2024. [Preserving multi-modal capabilities of pre-trained VLMs for improving vision-linguistic compositionality](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19060–19076. Association for Computational Linguistics.

Amit Peleg, Naman Deep Singh, and Matthias Hein. 2025. [Advancing compositional awareness in CLIP with efficient fine-tuning](#). In *Advances in Neural Information Processing Systems*, volume 38.

Hai X. Pham, David T. Hoffmann, Ricardo Guerrero, and Brais Martinez. 2026. [No hard negatives required: Concept centric learning leads to compositionality without degrading zero-shot capabilities of contrastive models](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8748–8763. PMLR.

A Matched Assignment Study: Complete Methods

A.1 Protocol and comparison

The matched assignment study follows the earlier twelve-policy experiment. Its scientific choices were frozen before its own fitting and scoring in `docs/REVIEW_FOLLOWUP_PROTOCOL.json`. It asks what label-informed positive assignment contributes beyond positive count, coarse frozen-score difficulty, and training schedule. The earlier protocol, outcomes, and inferential family remain separate. All four policy families use the same source captions and hypotheses, unit candidate weights, uniform positive targets, and the original symmetric mean-log-probability loss. The source policy makes only original captions positive. Supported promotion adds hypotheses labeled supported. The two randomized controls replace this added set, as defined below. Hypotheses that are not promoted remain candidates; they are not removed from the denominator. Original labels and evaluation targets are never overwritten.

A.2 Fixed promotion assignments

For training image i , let Q_i contain all its annotated hypotheses, S_i its source captions, and $H_i \subseteq Q_i$ its supported hypotheses. Write $K_i = |S_i|$ and $M_i = |H_i|$. The count-only control samples an M_i -element subset of Q_i uniformly without replacement. The score-stratified control first computes the cosine between each hypothesis and its owning image from cached frozen features, using explicit L2 normalization and float64 arithmetic. It orders hypotheses by increasing cosine, breaking exact ties by global manifest text index, and splits the ordered list into two bins with `numpy.array_split`; the lower-score bin receives the extra element for an odd count. In each bin, it uniformly samples exactly the number of hypotheses labeled supported originally in that bin. Thus the strata do not use labels, but their quotas do.

Each control has three independent assignment draws, seeded 101, 211, and 307 using NumPy

PCG64. Images are processed in manifest order. Each complete draw is fixed across all training seeds, learning rates, batches, and epochs. Empty and fully selected groups are retained. Naturally unchanged assignments are not redrawn. Distinct text IDs remain distinct candidates even when their strings coincide. Assignment records retain original labels, sampled IDs, label overlap, identity-level and exact-string changes, bin quotas, forced versus exchangeable groups, exact score ties, and frozen-score imbalance. Two bins match coarse score strata, not exact caption scores. Because their counts are label-derived, these controls are identification devices rather than annotation-free alternatives.

A.3 What the matched controls preserve

For any paired batch, all three expanded policies retain the same images, candidate IDs and strings, source positives, unit denominator weights, and $K_i + M_i$ positives per image. Each annotated text ID has exactly one owning image. Hence they also have the same number of eligible reverse anchors,

$$T_+ = \sum_{i=1}^B (K_i + M_i), \quad (2)$$

where B is the number of batch images. The identities of promoted hypothesis anchors can differ; their total count cannot. This holds for every batch, including the final partial batch.

A stronger fixed-score invariant follows. Let z_{ij} be the scaled image–text score, $p_{ij} = e^{z_{ij}} / \sum_k e^{z_{ik}}$ the image-row softmax, and $r_{ij} = e^{z_{ij}} / \sum_h e^{z_{hj}}$ the text-column softmax. Both sums use the unchanged complete candidate pools. Let P_i be the chosen positive captions for image i , E the eligible caption anchors, and $o(j)$ the owning image of eligible caption j . The full symmetric loss has derivative

$$\frac{\partial L}{\partial z_{ij}} = \frac{1}{2B} \left(p_{ij} - \frac{\mathbf{1}\{j \in P_i\}}{K_i + M_i} \right) + \frac{\mathbf{1}\{j \in E\}}{2T_+} (r_{ij} - \mathbf{1}\{i = o(j)\}), \quad (3)$$

where the second term is defined as zero for $j \notin E$. For any source-caption column, every quantity on the right is identical across supported, count-only, and score-stratified promotion at common logits. Their entire source-column score gradients therefore agree. The proof uses the unchanged source indicators, equal row-positive counts, equal

reverse-anchor counts, and unchanged softmax distributions. Gradients on other columns can differ. Shared adapter parameters combine all columns, so this identity does not imply identical parameter updates or source scores after training. It also does not compare expanded policies with source-only training, whose positive and reverse-anchor counts differ.

Expansion changes both directions. Within an image row, source targets change from $1/K_i$ to $1/(K_i + M_i)$. The reverse direction changes differently. Let T_0 be the number of original source-caption anchors, H the disjoint set of newly eligible hypothesis anchors, and ℓ_j^T a caption’s loss against the unchanged image pool. Each old caption retains its positive-image target and its individual loss at fixed scores, but

$$L_T^{\text{expanded}} = \frac{T_0 L_T^{\text{source}} + \sum_{j \in H} \ell_j^T}{T_0 + |H|}. \quad (4)$$

Thus original anchors retain fraction $T_0/(T_0 + |H|)$ of the reverse average. This is not an undiluted reverse objective. The fixed-score identities describe distinct task structures; they do not establish which change causes a trained model’s directional retrieval outcome.

A.4 Crossed fitting grid and replay

The original 1,200 training images and all their source captions and hypotheses are unchanged. Two deterministic policies and six randomized policy draws form eight fitting conditions. Crossing these with learning rates 10^{-4} , 3×10^{-4} , 10^{-3} and training seeds 17, 29, and 43 yields 72 fits. Assignment draws and training seeds are separate factors. Every condition uses the same image order for a given training seed, with batches of 32 images and ten epochs. Epochs 0 through 10 are retained, giving 792 state references. Epoch 10 defines the primary matched-schedule comparison; epochs 1 and 5 provide descriptive trajectories.

The frozen encoder, pretrained score scale, and zero-initialized independent 512×512 residual maps are inherited from the original study. AdamW uses weight decay .01, coefficients (.9, .999), epsilon 10^{-8} , gradient-norm clipping at one, and no scheduler; AMSGrad, fused, and foreach updates are disabled. Fitting uses deterministic CPU operations with two threads. Source and supported fitting call the unchanged original loss implementations.

The randomized masks call that same multipositive implementation after changing only training-positive assignments. All 18 source/supported candidates must reproduce the original training losses and native validation histories exactly, together with exact tensors at their original best epochs; a discrepancy stops execution. Immutable ledgers bind the protocol, original sources, features, six assignment files, and new code before fitting. Completion receipts bind every retained epoch state and validation artifact.

A.5 Source-caption retrieval pools

Source-only development retrieval uses the official e-ViL development image pool after excluding the original 100 calibration images: 900 images and 4,500 source captions. It includes the original 100 native-validation images. The test pool contains all 1,000 official e-ViL test images and their 5,000 source captions. Each image has five relevant source captions, and each caption has one relevant owning image. Hypotheses are absent from these evaluation candidate pools. Both directions search their complete pools. These are e-ViL-pool Flickr source-caption measurements, not the standard Karpathy Flickr1k split, which overlaps the original fitting and validation data.

Image IDs and image-file SHA256 values are checked for forbidden overlap across training, calibration, source development, and source test. The intentional inclusion of native validation within the development pool and of the original 400 relation-test images within the 1,000-image test pool is recorded. Source-caption provenance, repeated strings, and string overlap are audited separately. Cached features are reused only under matching IDs or exact content and encoder provenance. The original 400 test images and additional 600 images are reported as descriptive query subgroups using the same full candidate pool, not as separately selected benchmarks. The original VE400, Sugar-Crepe, SugarCrepe++, and COCO pools remain unchanged.

A.6 Two development selectors

The native selector exactly reuses the earlier criterion: the mean of source-or-supported image-to-text R@1 and image-balanced supported-versus-contradicted accuracy on the original 100 validation images and 1,875 caption IDs. Relation ties receive half credit. The source-retrieval selector averages image-to-text and text-to-image R@1 on

the source-only development pool above. For each policy or assignment draw and each learning rate, it chooses the best epoch separately for each training seed, including epoch 0; exact ties retain the earliest epoch. It then chooses one learning rate by the mean best score across the three training seeds, breaking ties toward the lower rate.

Each randomized draw is selected separately and the three draws receive equal weight in reporting. No best draw is selected. The two selectors define complete selection procedures: they differ in criterion, relevance, candidate pool, direction balance, and development-set size. Their comparison does not isolate the criterion alone. Selection references existing epoch states instead of duplicating weights. Reused states are scored once, and test outcomes do not enter checkpoint or learning-rate choice.

A.7 Evaluation and uncertainty

The primary family contains supported-minus-source, supported-minus-count-only, and supported-minus-score-stratified contrasts at each of three learning rates and in each source-retrieval R@1 direction, all at epoch 10: 18 effects. For a randomized comparator, first average its three assignment draws within each training seed and query, then average the paired seed differences. The three supported fits are reused as three fits, not treated as nine independent replications.

We use 10,000 paired image-cluster bootstrap resamples with seed 20261004 and Bonferroni-adjusted percentile intervals at nominal confidence $1 - .05/18 = 99.7222\%$ per contrast. Text-to-image resampling retains all five source-caption queries in their owning image cluster. These intervals condition on the fitted training seeds and assignment draws; neither factor is resampled. Query clusters are resampled against their original fixed retrieval gallery; gallery candidates are not regenerated. Per-seed and per-draw cells, their means, and all declared learning rates accompany the conditional image intervals. An interval spanning zero does not establish equivalence.

COCO, relation and composition tests, source metrics other than R@1, trajectories, subgroup summaries, and selected-procedure comparisons are descriptive. Source retrieval, VE400, SugarCrepe, and SugarCrepe++ cover the declared trajectory and selected states. COCO covers all epoch-10 states and unique selected states, without requiring additional epoch-1 or epoch-5 scoring. Adapter forward computation and normalization

remain float32; similarities accumulate in float64. Retrieval ties use candidate-manifest order, relation ties receive half credit, and composition triplets retain the original strict comparisons. Pre-score receipts bind code, input manifests and features, and checkpoint hashes. Per-query retrieval ranks and relevance indices, relation comparisons, and composition scores are retained for independent recomputation.

A.8 Interpretation and an unexecuted allocation design

The randomized controls test label-informed assignment beyond fixed counts and, for stratification, coarse frozen-score difficulty. They do not independently adjudicate visual truth. The directional outcomes establish behavior under the specified model and schedules; neither these outcomes nor Equation 4 identifies target redistribution as the cause of every earlier retrieval loss.

The archive also contains a separate, design-only allocation factorial. It proposes holding true positive identities fixed while changing source-versus-support image target mass and reverse-anchor group weights independently, with normalized total loss weight. This would test allocation directly. It has no implementation, fitted models, or results in this study and is outside the frozen protocol. The executed controls use only the original uniform-positive objective.

B Matched Assignment Study: Complete Results

B.1 Every terminal draw and training seed

Each row below is one epoch-ten fit. Count and Strata numbers identify assignment draws; no draw is selected for reporting. All entries are percentages.

B.2 Both complete selection strategies

B.3 Complete descriptive trajectory means

B.4 Prespecified trajectories

Rate	Comparator	$\Delta I \rightarrow T$	99.7222% interval	$\Delta T \rightarrow I$	99.7222% interval
10^{-4}	Source	-6.000	$[-9.037, -3.063]$	+0.120	$[-1.087, +1.322]$
10^{-4}	Count	-0.978	$[-3.389, +1.379]$	+1.811	$[+0.631, +2.995]$
10^{-4}	Strata	-0.744	$[-3.001, +1.436]$	+1.540	$[+0.504, +2.583]$
$3 \cdot 10^{-4}$	Source	-7.133	$[-10.367, -3.796]$	-1.320	$[-2.955, +0.381]$
$3 \cdot 10^{-4}$	Count	+1.322	$[-1.730, +4.114]$	+4.838	$[+3.332, +6.376]$
$3 \cdot 10^{-4}$	Strata	+0.122	$[-2.659, +2.815]$	+3.820	$[+2.412, +5.198]$
10^{-3}	Source	-10.833	$[-14.341, -7.330]$	-1.900	$[-3.962, +0.148]$
10^{-3}	Count	+3.789	$[+0.717, +6.922]$	+7.038	$[+5.163, +8.832]$
10^{-3}	Strata	+0.222	$[-2.591, +3.025]$	+4.084	$[+2.426, +5.804]$

Table 3: All 18 primary effects. Differences are supported minus comparator, in percentage points. Randomized comparators average all three assignment draws within each training seed before seed averaging. Intervals resample images and condition on the fitted seeds and draws.

Policy	Rate	Seed	Rel.	Edited	Both	e $I \rightarrow T$	e $T \rightarrow I$	C $I \rightarrow T$	C $T \rightarrow I$
Source	10^{-4}	17	77.99	81.01	63.93	84.90	65.74	53.98	36.95
Source	10^{-4}	29	77.62	80.93	64.31	84.60	66.12	53.82	36.97
Source	10^{-4}	43	77.65	81.04	64.39	84.60	66.06	53.52	36.97
Source	$3 \cdot 10^{-4}$	17	75.23	80.22	63.65	80.20	62.62	50.56	34.36
Source	$3 \cdot 10^{-4}$	29	75.62	79.96	63.61	82.30	63.00	50.74	34.63
Source	$3 \cdot 10^{-4}$	43	75.21	80.26	64.05	82.10	63.32	50.82	34.61
Source	10^{-3}	17	69.88	76.35	59.34	76.30	52.72	43.42	26.28
Source	10^{-3}	29	71.52	76.93	60.33	75.90	54.88	43.68	26.70
Source	10^{-3}	43	70.12	75.76	61.07	75.70	54.72	43.84	26.79
Supported	10^{-4}	17	87.64	83.96	67.90	78.40	65.96	49.02	36.54
Supported	10^{-4}	29	87.69	84.01	67.71	79.10	66.14	48.96	36.41
Supported	10^{-4}	43	87.52	83.84	67.63	78.60	66.18	49.16	36.35
Supported	$3 \cdot 10^{-4}$	17	86.54	83.26	66.93	74.90	61.08	45.06	33.38
Supported	$3 \cdot 10^{-4}$	29	86.79	83.54	66.91	74.40	62.16	45.24	33.82
Supported	$3 \cdot 10^{-4}$	43	86.71	83.25	67.63	73.90	61.74	43.46	33.70
Supported	10^{-3}	17	85.82	81.47	65.21	65.20	51.34	36.44	25.97
Supported	10^{-3}	29	85.90	81.33	65.38	65.70	52.96	37.40	26.62
Supported	10^{-3}	43	85.61	81.04	66.26	64.50	52.32	35.46	26.94

Table 4: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 1.

Policy	Rate	Seed	Rel.	Edited	Both	e $I \rightarrow T$	e $T \rightarrow I$	C $I \rightarrow T$	C $T \rightarrow I$
Count 1	10^{-4}	17	79.20	82.53	65.21	79.20	64.14	49.76	35.73
Count 1	10^{-4}	29	78.99	82.52	65.19	79.40	64.56	50.54	35.82
Count 1	10^{-4}	43	79.38	82.96	65.52	80.10	64.54	50.54	35.64
Count 1	$3 \cdot 10^{-4}$	17	76.45	80.55	63.19	72.10	56.52	44.30	31.06
Count 1	$3 \cdot 10^{-4}$	29	76.87	80.67	63.61	72.40	56.28	44.32	30.74
Count 1	$3 \cdot 10^{-4}$	43	75.85	79.79	63.63	73.30	56.54	44.04	30.42
Count 1	10^{-3}	17	74.92	77.82	59.81	61.20	45.12	34.88	22.57
Count 1	10^{-3}	29	74.66	78.15	61.57	60.00	43.60	33.94	22.14
Count 1	10^{-3}	43	73.58	77.41	60.04	63.70	43.96	34.80	22.04
Count 2	10^{-4}	17	78.90	82.89	64.91	79.90	63.98	49.76	35.55
Count 2	10^{-4}	29	79.07	82.76	65.65	78.30	64.52	49.82	35.47
Count 2	10^{-4}	43	79.20	82.93	65.15	79.70	64.28	50.12	35.48
Count 2	$3 \cdot 10^{-4}$	17	77.19	81.19	63.28	72.10	56.84	44.10	30.64
Count 2	$3 \cdot 10^{-4}$	29	75.93	81.05	63.55	73.40	56.66	44.22	30.56
Count 2	$3 \cdot 10^{-4}$	43	75.02	80.36	62.88	72.80	56.84	45.12	30.17
Count 2	10^{-3}	17	73.68	76.93	60.35	59.00	44.36	33.36	22.26
Count 2	10^{-3}	29	74.34	77.58	60.84	62.10	45.24	34.92	22.74
Count 2	10^{-3}	43	74.51	78.22	60.67	61.60	45.22	33.60	22.54

Table 5: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 1.

Policy	Rate	Seed	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Count 3	10^{-4}	17	79.26	82.65	64.98	79.80	64.18	49.98	35.15
Count 3	10^{-4}	29	78.75	82.45	64.98	80.60	64.14	50.18	35.14
Count 3	10^{-4}	43	79.05	82.61	64.83	80.10	64.20	50.14	35.29
Count 3	$3 \cdot 10^{-4}$	17	76.63	80.89	63.80	73.40	57.28	43.80	30.99
Count 3	$3 \cdot 10^{-4}$	29	75.82	80.69	63.53	74.00	56.96	44.18	30.59
Count 3	$3 \cdot 10^{-4}$	43	75.63	80.12	63.13	74.20	57.48	44.52	30.74
Count 3	10^{-3}	17	74.08	77.11	59.87	61.70	46.16	34.00	22.62
Count 3	10^{-3}	29	73.55	76.95	60.29	61.90	46.86	34.48	21.77
Count 3	10^{-3}	43	72.76	77.38	60.58	60.90	46.00	33.82	21.76
Strata 1	10^{-4}	17	81.51	82.80	65.82	78.90	64.44	50.18	35.71
Strata 1	10^{-4}	29	81.36	82.95	66.03	78.90	64.52	50.42	35.82
Strata 1	10^{-4}	43	80.98	82.98	65.92	78.90	64.64	49.88	35.70
Strata 1	$3 \cdot 10^{-4}$	17	78.88	81.47	63.88	74.00	57.64	45.44	31.31
Strata 1	$3 \cdot 10^{-4}$	29	78.19	81.15	63.95	72.30	58.22	45.42	31.26
Strata 1	$3 \cdot 10^{-4}$	43	78.24	81.07	64.70	74.10	58.52	45.46	31.38
Strata 1	10^{-3}	17	76.90	79.23	62.94	65.20	45.86	36.06	23.48
Strata 1	10^{-3}	29	76.16	77.50	62.10	63.70	49.74	37.00	24.13
Strata 1	10^{-3}	43	75.27	79.34	62.06	63.50	47.22	35.84	23.41

Table 6: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 1.

Policy	Rate	Seed	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Strata 2	10^{-4}	17	80.71	83.17	65.84	79.90	64.72	50.30	35.74
Strata 2	10^{-4}	29	81.00	83.06	66.47	79.50	64.78	50.14	35.74
Strata 2	10^{-4}	43	80.49	83.53	66.07	80.10	64.46	49.80	35.65
Strata 2	$3 \cdot 10^{-4}$	17	78.51	82.71	65.55	73.60	57.72	44.82	31.58
Strata 2	$3 \cdot 10^{-4}$	29	78.18	82.01	65.21	75.30	57.18	45.66	31.54
Strata 2	$3 \cdot 10^{-4}$	43	77.73	81.85	65.10	73.80	58.58	45.62	31.52
Strata 2	10^{-3}	17	76.49	79.10	62.73	64.80	47.82	36.80	23.46
Strata 2	10^{-3}	29	75.84	79.55	62.37	66.70	49.52	38.30	23.83
Strata 2	10^{-3}	43	76.65	79.56	62.98	65.20	48.40	37.00	24.16
Strata 3	10^{-4}	17	81.49	82.64	65.38	79.40	64.54	50.10	35.50
Strata 3	10^{-4}	29	81.24	82.93	65.57	79.90	64.54	50.14	35.62
Strata 3	10^{-4}	43	81.38	82.95	65.76	79.50	64.34	49.88	35.46
Strata 3	$3 \cdot 10^{-4}$	17	78.48	81.51	63.53	74.80	57.60	45.50	31.18
Strata 3	$3 \cdot 10^{-4}$	29	78.14	81.25	63.72	74.30	57.64	45.82	30.96
Strata 3	$3 \cdot 10^{-4}$	43	78.60	81.43	63.72	76.30	57.46	45.68	31.30
Strata 3	10^{-3}	17	76.59	79.06	61.05	63.70	47.60	36.26	23.56
Strata 3	10^{-3}	29	76.18	77.95	62.10	66.30	48.70	37.30	23.81
Strata 3	10^{-3}	43	76.86	79.11	62.14	65.10	48.24	35.02	24.02

Table 7: Terminal per-seed results. e denotes the full e-ViL source-caption pool and C denotes COCO. Other endpoints follow Table 1.

Selector	Policy	Seed	Ep.	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Native	Source	17	1	81.35	83.44	67.65	84.90	69.68	56.28	39.65
Native	Source	29	1	81.09	83.42	68.09	85.00	69.74	56.02	39.46
Native	Source	43	3	81.12	82.91	66.47	84.20	68.64	55.26	39.02
Native	Supported	17	5	86.75	83.38	67.58	78.20	63.62	48.36	35.40
Native	Supported	29	10	86.79	83.54	66.91	74.40	62.16	45.24	33.82
Native	Supported	43	5	87.46	83.25	67.19	78.40	64.12	48.28	35.30
Native	Count 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 3	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Count 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35

Table 8: Every selected training-seed state. Each random draw has its own learning-rate and epoch selection; there is no winning-draw search. The source manifest records exact state IDs and learning rates.

Selector	Policy	Seed	Ep.	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Native	Strata 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 3	17	7	80.11	82.09	65.00	78.30	60.84	48.36	33.71
Native	Strata 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Native	Strata 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Source	17	1	81.35	83.44	67.65	84.90	69.68	56.28	39.65
Retrieval	Source	29	2	80.96	83.29	67.06	84.90	69.64	56.04	39.60
Retrieval	Source	43	2	81.15	83.14	67.08	84.70	69.78	55.94	39.40
Retrieval	Supported	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Supported	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Supported	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35

Table 9: Every selected training-seed state. Each random draw has its own learning-rate and epoch selection; there is no winning-draw search. The source manifest records exact state IDs and learning rates.

Selector	Policy	Seed	Ep.	Rel.	Edited	Both	e I→T	e T→I	C I→T	C T→I
Retrieval	Count 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 3	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Count 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 1	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 1	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 1	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 2	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 2	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 2	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 3	17	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 3	29	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35
Retrieval	Strata 3	43	0	79.47	81.21	68.95	84.50	69.76	56.32	39.35

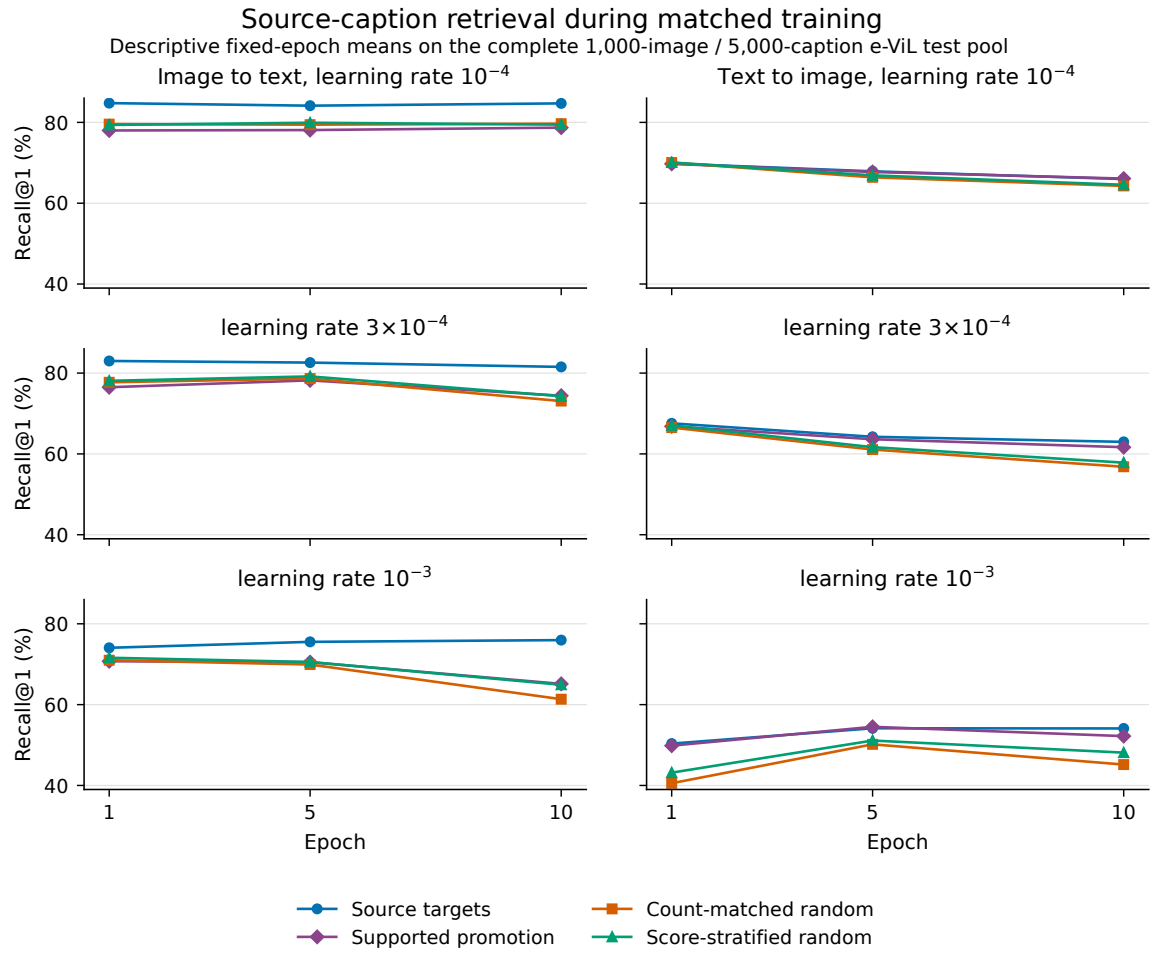
Table 10: Every selected training-seed state. Each random draw has its own learning-rate and epoch selection; there is no winning-draw search. The source manifest records exact state IDs and learning rates.

Policy	Rate	Ep.	Rel.	Edited	Both	First	Alternative	e I→T	e T→I
Source	10^{-4}	1	81.23	83.40	67.94	80.95	73.10	84.77	69.81
Source	10^{-4}	5	80.07	81.89	65.55	80.16	70.25	84.13	67.89
Source	10^{-4}	10	77.75	81.00	64.21	80.16	68.78	84.70	65.97
Source	$3 \cdot 10^{-4}$	1	80.63	82.32	65.48	80.22	70.38	83.00	67.57
Source	$3 \cdot 10^{-4}$	5	76.53	80.17	63.55	79.41	68.38	82.60	64.24
Source	$3 \cdot 10^{-4}$	10	75.35	80.14	63.77	79.22	68.63	81.53	62.98
Source	10^{-3}	1	74.94	77.38	61.03	77.34	65.76	74.07	50.36
Source	10^{-3}	5	73.25	77.57	61.71	77.35	67.05	75.53	54.15
Source	10^{-3}	10	70.51	76.35	60.25	76.59	65.51	75.97	54.11
Supported	10^{-4}	1	86.72	85.43	70.00	81.18	75.69	78.00	69.74
Supported	10^{-4}	5	87.91	84.58	68.25	80.76	73.58	78.10	67.73
Supported	10^{-4}	10	87.62	83.93	67.75	80.23	73.31	78.70	66.09
Supported	$3 \cdot 10^{-4}$	1	87.48	84.82	68.47	80.42	74.13	76.50	66.83
Supported	$3 \cdot 10^{-4}$	5	87.04	83.34	67.33	79.81	72.90	78.20	63.63
Supported	$3 \cdot 10^{-4}$	10	86.68	83.35	67.16	79.45	72.84	74.40	61.66
Supported	10^{-3}	1	86.17	81.61	65.42	78.12	71.93	70.73	49.85
Supported	10^{-3}	5	86.03	81.66	66.97	78.01	73.22	70.50	54.51
Supported	10^{-3}	10	85.78	81.28	65.62	77.37	71.94	65.13	52.21

Table 11: All declared descriptive epoch means. First and Alternative are the separate valid-caption SugarCrepe++ accuracies. Random families average all three assignments equally within seed.

Policy	Rate	Ep.	Rel.	Edited	Both	First	Alternative	e I→T	e T→I
Count mean	10^{-4}	1	83.71	84.56	69.80	80.83	75.53	79.57	70.06
Count mean	10^{-4}	5	81.39	83.49	66.13	79.83	71.24	79.43	66.40
Count mean	10^{-4}	10	79.09	82.70	65.16	79.62	70.56	79.68	64.28
Count mean	$3 \cdot 10^{-4}$	1	82.61	83.78	66.97	79.60	72.37	77.72	66.52
Count mean	$3 \cdot 10^{-4}$	5	77.88	81.98	64.02	79.18	69.17	78.64	61.08
Count mean	$3 \cdot 10^{-4}$	10	76.15	80.59	63.40	77.89	69.25	73.08	56.82
Count mean	10^{-3}	1	77.54	80.55	63.60	77.71	69.64	70.99	40.55
Count mean	10^{-3}	5	74.76	79.16	62.78	76.54	69.00	69.92	50.18
Count mean	10^{-3}	10	74.01	77.51	60.45	75.14	66.54	61.34	45.17
Strata mean	10^{-4}	1	84.19	84.61	69.84	80.82	75.73	79.37	70.06
Strata mean	10^{-4}	5	82.86	83.67	66.90	80.06	72.19	79.91	66.89
Strata mean	10^{-4}	10	81.13	83.00	65.87	79.75	71.35	79.44	64.55
Strata mean	$3 \cdot 10^{-4}$	1	83.65	83.84	67.52	79.70	73.11	78.09	67.06
Strata mean	$3 \cdot 10^{-4}$	5	80.18	82.38	65.28	79.28	70.88	79.18	61.67
Strata mean	$3 \cdot 10^{-4}$	10	78.33	81.60	64.37	78.54	70.09	74.28	57.84
Strata mean	10^{-3}	1	79.54	80.50	64.05	77.54	70.21	71.60	43.17
Strata mean	10^{-3}	5	77.59	79.91	64.49	77.04	70.70	70.56	51.11
Strata mean	10^{-3}	10	76.32	78.93	62.27	76.27	68.27	64.91	48.12

Table 12: All declared descriptive epoch means. First and Alternative are the separate valid-caption SugarCrepe++ accuracies. Random families average all three assignments equally within seed.



Every planned learning rate and epoch is shown. Three training seeds; randomized policies average three fixed assignment draws.

Figure 2: Descriptive source-pool retrieval at epochs one, five, and ten. Every learning rate is displayed. Randomized families average all assignment draws equally within seed. The primary family concerns epoch ten only.

C Prior Study: Context and Selected Procedures

The following appendices preserve the completed twelve-policy study that motivated the matched-assignment experiment. Its fits, independently selected procedures, six-effect inferential family, and 400-image relation pool are distinct from the follow-up’s matched terminal contrasts and full source-caption retrieval pool. Its tables do not estimate the effect of positive assignment at a common training schedule. All source files and results of that study are retained unchanged in the project archive; this appendix changes only presentation and cross-references.

The selected-procedure comparison used one common tuning budget per rule. The complete policy definitions, source provenance, per-seed outcomes, diagnostic, and mathematical analysis fol-



Source positives: A young girl is preparing to blow out her candle.

Positive expansion: a woman is making some food in a bowl

Figure 3: Top retrieved captions for the same image query (seed 17). The best annotated caption moves from rank 1 to 6. Both models search the full caption pool.

low. The caption-retention figure below concerns only the original selected states; the follow-up retains additional early states and reports their trajectories separately.

D Prior Study: Complete Policy Definitions

A batch has B unique image rows and T text columns, containing all selected source and hypothesis captions for its images. The relation matrix $R \in \{0, 1, 2, 3, 4\}^{B \times T}$ uses codes for unannotated, source, supported, contradicted, and neutral pairs, respectively. Relations apply only to their annotated image–text pair. An absent cross-image annotation is an eligible negative, not a neutral label.

Let J denote the candidate set for a row or column, $P \subseteq J$ its positive set, and P_{ij} the binary positive mask. Every contrastive policy includes source captions. Expanded policies additionally include supported hypotheses. Candidate weights start at one and are changed according to Table 16; all positive weights are then forced to one. The scalar loss averages eligible image rows and eligible text columns independently, with weight $1/2$ on each direction. The reverse direction transposes the same mask and weight matrices. Columns without a positive image are omitted from the reverse average. These rules preserve the meaning of rectangular batches with several captions per image.

Detached similarity weights. Let t_j be the unit frozen text feature for candidate j , and C_i the

Policy	e-SNLI-VE	SugarCrepe	SC++ both	COCO I→T	COCO T→I
Frozen	79.47	81.21	68.95	56.32	39.35
Source positives	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40	55.85	39.38
Positive expansion	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23	47.29	34.84
+ contradiction weighting	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30	47.25	34.81
+ neutral exclusion	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66	48.01	35.39
+ both	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61	47.26	35.27
Constant weights	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37	55.21	39.39

Table 13: Relation interventions with the same candidate pool and tuning budget. Scores are percentages; subscripts show sample standard deviation across three selected seeds for the primary endpoints. COCO columns report full-pool R@1. The three rows marked + add rules to positive expansion. Constant weights use source positives.

Edit group	n	Source	Expand.	Δ
Added content	2,754	88.09	90.01	+1.92
Replace/swap	4,757	80.46	79.56	-0.90
All edits	7,511	83.26	83.39	+0.14

Table 14: SugarCrepe accuracy by edit group. Differences are positive expansion minus source positives, in percentage points. Groups partition the official pairs.

Weights	e-SNLI-VE	Sugar	I→T
Fixed similarity	79.56	81.80	54.33
Shuffled weights	83.34	83.49	52.27
Median similarity	82.66	83.73	53.97

Table 15: Similarity weights and their shuffle, using source positives. Entries are mean percentages; the final column is COCO R@1. The constant-weight reference appears in Table 13.

source-caption indices of image i . Define

$$a_{ij} = \frac{1 + |C_i|^{-1} \sum_{k \in C_i} t_k^\top t_j}{2}. \quad (5)$$

The average of source features is not renormalized. The negative weight is

$$w_{ij} = \text{sigmoid}\left(\frac{\tau - a_{ij}}{0.05}\right). \quad (6)$$

The fixed rule sets $\tau = 0.5$. The median rule uses the statistical median of a_{ij} over nonpositive cells of the current batch, averaging the two central values for an even count. Positive weights remain one. All features, thresholds, and weights in this calculation are detached from the training graph. The kernel never uses adapted features.

Shuffling and random exclusion. The shuffled rule permutes the complete batch-wide multiset of negative weights from the fixed-threshold rule, leaving positive weights intact. It preserves the weight distribution but need not preserve each row’s weighted mass. Random exclusion samples nonpositive cells without replacement within each image row, matching that image’s number of neutral annotations. Because the reverse direction uses a transpose, its per-text exclusion counts need not match the neutral rule. Both randomizations use a seeded generator whose state advances through batches and epochs.

Smoothing and ranking. Smoothing uses source-positive targets and unit weights. Its row target is $0.9 \text{Unif}(P) + 0.1 \text{Unif}(J)$, where J contains every candidate. For ranking, let E_i and C'_i denote supported and contradicted hypotheses of image i . The eligible-image loss is

$$\ell_i^{\text{rank}} = \frac{\sum_{e \in E_i} \sum_{c \in C'_i} \text{softplus}(s_{ic} - s_{ie})}{|E_i| |C'_i|}. \quad (7)$$

We average over images with both sets nonempty. Ranking has no reverse direction and does not train on source, neutral, or unannotated pairs.

E Prior Study: Data, Provenance, and Reconstruction

The current execution was reconstructed from retained methods and proof text after the earlier workspace became unavailable. Historical aggregate tables are kept in a separate context directory and are not inputs to training, evaluation, or paper tables. The exact old similarity kernel and the base target set for smoothing were not recoverable. The explicit definitions in Appendix D were fixed

Archive name	Positive set	Weight or target rule
clip	Source	All weights one
multipositive	Source + supported	All weights one
grounded	Source + supported	Contradictions: 2; neutrals: 0
grounded_no_hardenening	Source + supported	Neutrals: 0
grounded_no_abstention	Source + supported	Contradictions: 2
random_exclusion	Source + supported	Count-matched random exclusions
smoothing	Source	$0.9 \text{ Unif}(P) + 0.1 \text{ Unif}(J)$
constant	Source	All negatives: 0.25
sanw_fixed	Source	Similarity kernel, threshold 0.5
sanw_median	Source	Similarity kernel, batch median threshold
shuffled	Source	Permuted fixed-threshold negative weights
pairwise_rank	Supported/contradicted	Image-balanced pairwise softplus

Table 16: All twelve trainable policies. A weight of zero exactly excludes a pair. The frozen reference has no trained parameters.

Split	Images	Support	Contradict	Neutral
Train	1,200	5,319	5,846	5,027
Calibration	100	522	554	355
Validation	100	502	534	339
Test	400	2,088	2,278	1,495

Table 17: Retained hypothesis annotations. Each image also has five source captions. All validation and test images, and 1,192 of 1,200 training images, have both supported and contradicted hypotheses.

before the new runs and are documented in the reconstruction record. The prior twelve-policy study was a new execution under those recorded choices. The matched-assignment follow-up preserves those original inputs and outputs.

E.1 Relation annotations and source captions

We use the official e-ViL release of e-SNLI-VE at revision 94f17f5c1895. Its training, development, and test sources contain 401,717, 14,339, and 14,740 annotated hypotheses over 29,783, 1,000, and 1,000 images, respectively. The official pools have no shared image IDs or annotation-pair IDs. With Python’s `random.Random(42)`, we sample from sorted image IDs: 1,200 training images, 200 development images split into 100 calibration and 100 validation images, and 400 test images. All annotations of each selected image remain in source order. Table 17 gives the resulting counts.

The source captions come from the official Flickr30k Entities annotation archive at revision 68b3d6f12d1d. We use its five sentence-file

lines per image in stored order. Entity markup is removed while retaining source words, case, and punctuation, with single spaces between stored tokens. This source covers all 31,783 images. An incomplete alternative caption mirror covers only 31,014 and was not used. The image bytes come from the `nlphuji/flickr30k` mirror at revision 2b239befc81b. Each selected image passes ZIP size/CRC checks, JPEG verification, and full decoding. No selected image is silently skipped.

An earlier deprecated relation release was rejected during an input-only audit because of development/test overlap. This decision preceded feature-based evaluation. The chosen official release, exact IDs, source URLs, full revisions, file hashes, and selection order are recorded in the per-dataset provenance files. The released labels include inherited text-inference judgments, so their names are operational annotation categories rather than a claim of new visual adjudication.

E.2 Transfer benchmarks

SugarCrepes uses all 7,511 pairs over 1,560 images from revision 0047054b2439. Its category counts are 692 added attributes, 2,062 added objects, 788 replaced attributes, 1,652 replaced objects, 1,406 replaced relations, 666 swapped attributes, and 245 swapped objects. SugarCrepes++ uses all 4,757 triplets over 1,542 images from revision c083bbf039a5; it covers the five replacement/swap categories with the same category counts.

Each benchmark preserves its own released text

strings. SugarCrepe++ changes 1,203 first-positive strings and 878 foil strings relative to the matched SugarCrepe records. Its first-positive accuracy therefore cannot substitute for SugarCrepe’s replacement/swap score. Two released SugarCrepe++ triplets have identical first-positive and foil strings, making a strict win impossible; these remain in the official denominator. One triplet repeats the same string in both positive positions. No item is edited or removed in response to model scores.

COCO uses all 5,000 Stanford Karpathy test images. The source contains 25,010 captions; taking the first five in stored order for every image yields 25,000 candidates and omits ten sixth captions. Caption strings are not normalized. The COCO JPEG cache contains 6,377 distinct benchmark images. It uses the official `val2014` source for Karpathy members and `val2017` for remaining benchmark images. All 183 shared SugarCrepe/Karpathy images were checked across these source versions and are byte-identical.

E.3 Overlap and duplicate audit

All 1,542 SugarCrepe++ images also occur in SugarCrepe. COCO retrieval shares 183 images with SugarCrepe and 182 with SugarCrepe++. These are complementary measurements on overlapping pools. The selected Flickr30k images share neither IDs nor image-byte hashes with the COCO-derived benchmarks. The four e-SNLI-VE splits also have no shared image bytes.

The input audit finds 48 repeated exact image/text groups in e-SNLI-VE, each appearing twice. Of these, 46 are repeated hypothesis groups. Their relation labels do not conflict. All rows are retained, so repeated annotations retain their multiplicity in the loss and metrics. The two positive/foil equality cases account for the within-triplet relation conflicts in SugarCrepe++. The audit records source defects before scoring rather than changing benchmark contents.

E.4 Frozen feature extraction

The encoder is laion/CLIP-ViT-B-32-laion2B-s34B-b79K, revision 1a25a446712b, with weight SHA256 beginning ac4f8c4b88af. We use OpenCLIP 2.32.0 and PyTorch 2.7.1 on CPU. Images are resized with bicubic interpolation, center-cropped to 224×224 , converted to RGB, and normalized with the checkpoint’s channel means and standard deviations. The 512-dimensional image and text outputs

are stored in float32 after unit L2 normalization. The fixed scale is 100.00000762939453.

Text encoding trims only causal padding positions after the last end-of-text token in a batch. A parity check compares normalized outputs against the stock encoder with tolerance $2 \cdot 10^{-6}$. Recorded per-dataset parity checks, preprocessing parameters, source hashes, input-manifest hashes, feature-cache hashes, and full model pins are stored in each feature metadata file. Reused features require exact image-byte or text-string identity under the same encoder and preprocessing.

F Prior Study: Training and Checkpoint Selection

Let $x, z \in \mathbb{R}^{512}$ be unit frozen image and text features. The trainable maps are

$$\hat{x} = \frac{x + A_I x}{\|x + A_I x\|_2}, \quad \hat{z} = \frac{z + A_T z}{\|z + A_T z\|_2}. \quad (8)$$

Both 512×512 matrices start at zero and have no bias. Their total parameter count is $2 \cdot 512^2 = 524,288$. The score is $s_{ij} = \alpha \hat{x}_i^\top \hat{z}_j$, where $\alpha > 0$ is the pretrained checkpoint’s fixed scale. Feature extraction and source hashes identify the exact encoder and input order.

The learning-rate grid is $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ and training seeds are 17, 29, 43. Each candidate trains for ten epochs with image batches of 32. All source captions and hypotheses assigned to those images enter the text candidate pool. AdamW uses weight decay 0.01, betas (0.9, 0.999), and epsilon 10^{-8} . The gradient norm is clipped at one. There is no warmup or learning-rate schedule. Separate random generators control image order and stochastic policy operations, so weight shuffling does not alter batch order.

Validation retrieval uses every validation caption and hypothesis as a candidate. Both source captions and supported hypotheses are relevant matches. Its R@1 and the image-balanced supported-versus-contradicted accuracy receive equal weight in the selection score. The epoch with the highest score is retained for each seed and learning rate, including epoch zero. Exact ties retain the earliest epoch. A policy selects one learning rate by averaging these best scores over all three seeds; exact ties select the smaller rate. Held-out scoring starts after all 108 candidates and 36 selections are frozen.

Ranking evaluation accumulates dot products in float64 over adapted float32 normalized features.

Exact retrieval-score ties follow candidate manifest order; relation-ranking ties receive half credit. The fixed positive score scale is omitted from ranking metrics because it cannot change the order. Images without both supported and contradicted hypotheses are omitted only from the relation average, with their count recorded.

An immutable execution ledger binds the protocol, code, manifest, feature metadata, cache, and split IDs before fitting. Every candidate saves its best state, all eleven validation evaluations, and a completion record with artifact hashes. The root epoch-zero state is retained separately. Resume accepts a completed candidate only when its identities and hashes match. Interrupted candidates restart deterministically from epoch zero. Selected checkpoints are copied into a separate tree while all candidate states remain available.

G Prior Study: Metrics and Statistical Comparisons

For e-SNLI-VE, compare every supported hypothesis against every contradicted hypothesis within an image. A larger supported score receives one point and an exact tie receives half a point. Average pairs within each eligible image, then average images. This gives each image equal weight despite different numbers of annotated statements.

SugarCrepe uses a strict positive-score comparison against its foil. SugarCrepe++ reports each positive separately and their conjunction against the same foil. The conjunction is one only if both scores are strictly greater. Per-category results retain official labels. The addition group pools added objects and attributes; the combined replacement/swap group pools the remaining categories. Grouped scores use item counts as weights, not an unweighted mean of category percentages.

COCO image-to-text R@1 counts an image query as correct when its highest-scoring caption belongs to the image’s annotated caption set. Text-to-image R@1 requires the source image at rank one. All candidates remain in the full pool during scoring. Source-caption ownership is the operational relevance label.

The six primary contrasts compare the combined relation policy against positive expansion, neutral exclusion alone, and contradiction weighting alone, on e-SNLI-VE and SugarCrepe. For each contrast, average the paired outcomes across the three selected seeds before sampling images. Each boot-

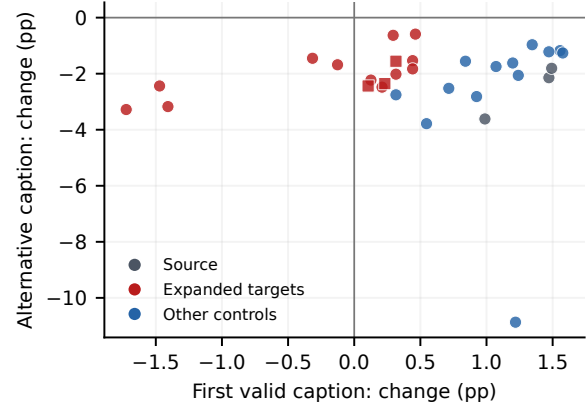


Figure 4: Changes on the same SugarCrepe++ triplets relative to frozen. Each point is one selected nonzero-epoch adapter. All 31 original selected adapters lose accuracy on the alternative valid caption; 26 improve on the first. The 5 epoch-zero selections are omitted.

strap draw samples images with replacement and retains all associated items. With 10,000 replicates, seed 20261003, and family size six, two-sided percentile bounds use tail probability $0.05/(2 \cdot 6)$, yielding 99.1667% intervals. They condition on the selected runs and data images. Secondary metrics and category groupings are descriptive.

H Prior Study: Retention Across Selected Adapters

The first-positive and alternative-positive scores use the same triplets and foil. The decline in alternative-caption accuracy is relative to frozen and is not specific to positive expansion. Positive expansion has slightly higher alternative-caption accuracy than source-positive adaptation. Full per-policy and per-seed values follow below.

I Prior Study: Full Fresh Results and Execution Checks

Tables 18–25 give all selected-policy means, seed outcomes, primary intervals, category scores, and retrieval transitions. Five selected states retain epoch zero: all three ranking runs and fixed-similarity seeds 29 and 43. Those states have predictions identical to the frozen reference.

The illustrative query in Figure 3 is coco 60623. It is the first image in COCO manifest order that source-positive adaptation retrieves correctly at rank one in all three seeds, while positive expansion fails at rank one in every seed and falls outside the top five in seed 17. The displayed captions are the two seed-17 top results. This deterministic post

Policy	e-SNLI-VE	SugarCrepe	SC++ both
Frozen	79.47	81.21	68.95
Source	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40 $\pm$ 0.84
Expanded	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23 $\pm$ 0.34
Contradiction	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30 $\pm$ 0.28
Neutral	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66 $\pm$ 0.52
Combined	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61 $\pm$ 0.45
Random	86.26 $\pm$ 0.28	81.62 $\pm$ 0.70	65.93 $\pm$ 0.57
Smoothing	79.23 $\pm$ 4.85	82.28 $\pm$ 2.18	65.97 $\pm$ 4.59
Ranking	79.47 $\pm$ 0.00	81.21 $\pm$ 0.00	68.95 $\pm$ 0.00
Fixed	79.56 $\pm$ 0.15	81.80 $\pm$ 1.01	68.00 $\pm$ 1.65
Shuffled	83.34 $\pm$ 0.57	83.49 $\pm$ 0.36	66.90 $\pm$ 0.26
Median	82.66 $\pm$ 0.19	83.73 $\pm$ 0.17	68.01 $\pm$ 0.07
Constant	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37 $\pm$ 0.48

Table 18: All selected policies on the three discrimination endpoints. Entries are percentages with sample seed standard deviations. Source, Expanded, Contradiction, Neutral, and Combined correspond to the first five trainable policies in Table 13. Other names follow Table 16.

hoc choice illustrates a recorded failure and is not a representative sample or an extra significance test.

J Separate Relation-Estimator Diagnostic

This diagnostic predicts released supported, contradicted, and neutral labels from frozen features. It does not provide labels or weights to the adapter study. Source captions and unannotated cross-image pairs are excluded. Each annotated image/hypothesis pair is one example, and each image remains entirely within its training, validation, calibration, or test split.

Features and fitting. The image-only and text-only models use the corresponding 512-dimensional features. The multimodal model concatenates image, text, their elementwise product, and their absolute difference. A standard scaler is fitted only on training rows. Multinomial logistic regression uses an intercept, L2 regularization, `lbfgs`, tolerance 10^{-6} , no class weights, and seed 17. The grid $C \in \{0.01, 0.1, 1, 10\}$ is selected by validation macro-F1, with smaller C breaking ties. There is no refit on validation data.

The original reconstruction set a 2,000-iteration cap. Text fitting reached that cap, so a separately recorded amendment raised the cap to 10,000 without changing the model, grid, splits, or tolerance. The amended run has its own ledger and output directory; the original diagnostic remains archived. This numerical-completion amendment does not

alter the frozen primary-adapter protocol.

Calibration and selection gates. Sort calibration image IDs and permute them with NumPy seed zero. The first 50 images fit a positive scalar temperature by minimizing mean three-class negative log likelihood; the other 50 select gates. Temperature lies in $[0.05, 20]$, optimized in log space, with one and both endpoints also considered explicitly. It preserves the classifier’s argmax predictions.

For supported and contradicted predictions separately, a gate requires that class to be the argmax and its calibrated probability to meet or exceed a threshold. Search complete confidence-tie groups and choose maximum coverage among thresholds accepting at least 20 calibration pairs with a two-sided 95% Wilson lower endpoint of at least 0.90. If no threshold qualifies, the gate accepts nothing. This calibration criterion is not a simultaneous guarantee after threshold search. Final precision and coverage are measured on held-out test images. Coverage divides accepted count by all test hypotheses; precision divides correct accepted predictions by accepted count. An empty gate has undefined precision.

Metrics and out-of-fold predictions. The diagnostic reports accuracy, three-class macro-F1, 10-bin expected calibration error, Brier score, and negative log likelihood. Brier score sums squared error over the three classes before averaging examples. Calibration bins have equal width and include probability one in the last bin. Class ties follow

Policy	SC++ first	SC++ alternative	COCO I→T	COCO T→I
Frozen	79.46	75.01	56.32	39.35
Source	80.78 $\pm$ 0.29	72.48 $\pm$ 0.96	55.85 $\pm$ 0.53	39.38 $\pm$ 0.32
Expanded	79.68 $\pm$ 0.11	72.89 $\pm$ 0.49	47.29 $\pm$ 1.78	34.84 $\pm$ 0.89
Contradiction	79.72 $\pm$ 0.16	72.92 $\pm$ 0.49	47.25 $\pm$ 1.94	34.81 $\pm$ 0.92
Neutral	79.72 $\pm$ 0.33	73.64 $\pm$ 0.68	48.01 $\pm$ 0.15	35.39 $\pm$ 0.23
Combined	79.56 $\pm$ 0.36	73.64 $\pm$ 0.70	47.26 $\pm$ 1.17	35.27 $\pm$ 0.44
Random	77.93 $\pm$ 0.17	72.04 $\pm$ 0.46	43.43 $\pm$ 1.23	29.95 $\pm$ 0.65
Smoothing	80.88 $\pm$ 0.18	70.56 $\pm$ 5.56	53.85 $\pm$ 2.54	37.08 $\pm$ 3.15
Ranking	79.46 $\pm$ 0.00	75.01 $\pm$ 0.00	56.32 $\pm$ 0.00	39.35 $\pm$ 0.00
Fixed	79.64 $\pm$ 0.32	73.74 $\pm$ 2.18	54.33 $\pm$ 3.45	36.86 $\pm$ 4.31
Shuffled	80.11 $\pm$ 0.31	72.31 $\pm$ 0.15	52.27 $\pm$ 0.29	38.19 $\pm$ 0.12
Median	80.50 $\pm$ 0.18	73.37 $\pm$ 0.10	53.97 $\pm$ 0.40	39.21 $\pm$ 0.27
Constant	80.84 $\pm$ 0.16	73.60 $\pm$ 0.58	55.21 $\pm$ 0.14	39.39 $\pm$ 0.09

Table 19: Valid-caption and retrieval retention. Entries are percentages with sample seed standard deviations.

the fixed order supported, contradicted, neutral. Nonempty gate precision receives a descriptive 95% image-bootstrap interval from 10,000 replicates with seed 20261003. Empty-acceptance replicates are counted and omitted from the percentile calculation.

For reusable training predictions, sort training image IDs, permute with seed 17, and divide them into three image-disjoint folds. Each fold trains on the other two folds, selects C on the external validation split, and calibrates on the external calibration halves. Thus held-out training labels do not enter the model or selection decisions producing their predictions. These predictions are archived for future work and do not enter adapter fitting.

K Prior Study: Reproducibility and Extensions

The archive separates source acquisition, normalized feature extraction, fit/selection, held-out prediction, statistical analysis, and manuscript generation. Manifests give stable IDs and input hashes. Candidate checkpoints and validation histories establish which states were eligible for selection. Selected predictions support metric reconstruction without rerunning training. Result tables are generated from the fresh analysis artifact rather than entered from the historical manuscript.

Extensions retained for future work include a second backbone, full-encoder training, more training data and seeds, independently image-adjudicated relations, denominator-matched and exact gradient-mass-matched training controls, hardness-matched

exclusion, the summed-positive objective, predicted relations, inverse-multiplicity weighting, retention-constrained selection, feature distillation, many-to-many retrieval relevance, natural deployment queries, and frozen retrieval followed by relation-adapter reranking. A reranker can change the top result even when its candidate generator is fixed. Matching controls must specify the image and text directions separately. These are distinct from the completed comparisons reported in the fresh results.

L Prior Study: Analytical Identities and Their Scope

This appendix preserves the prior study’s positive-target identities and complete negative-weight analysis. Unless an alternative is specified, the statements concern the unsmoothed uniform-positive objective at specified scores. Their proofs do not depend on experimental outcomes.

L.1 Notation and the common gradient

Let J be a finite candidate set, $P \subseteq J$ a nonempty positive set, and $N = J \setminus P$. Each score $s_j \in \mathbb{R}$ is finite. Weights $w_j \geq 0$ are fixed for differentiation and satisfy $w_p = 1$ for $p \in P$. Define

$$D = \sum_{j \in J} w_j e^{s_j}, \quad q_j = \frac{w_j e^{s_j}}{D}, \quad (9)$$

$$y_j = \frac{\mathbf{1}\{j \in P\}}{|P|}, \quad \ell = \log D - \sum_j y_j s_j. \quad (10)$$

Since P is nonempty and scores are finite, $D > 0$. A zero weight removes a candidate from the

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Source	17	0.0001	1	81.35	83.44	67.65	56.28	39.65
Source	29	0.0001	1	81.09	83.42	68.09	56.02	39.46
Source	43	0.0001	3	81.12	82.91	66.47	55.26	39.02
Expanded	17	0.0003	5	86.75	83.38	67.58	48.36	35.40
Expanded	29	0.0003	10	86.79	83.54	66.91	45.24	33.82
Expanded	43	0.0003	5	87.46	83.25	67.19	48.28	35.30
Contradiction	17	0.0003	5	87.04	83.33	67.58	48.46	35.48
Contradiction	29	0.0003	10	87.13	83.49	67.02	45.02	33.76
Contradiction	43	0.0003	5	87.66	83.17	67.31	48.28	35.18
Neutral	17	0.0003	5	86.42	83.60	68.26	48.10	35.54
Neutral	29	0.0003	5	86.40	83.49	67.35	47.84	35.50
Neutral	43	0.0003	5	87.17	83.09	67.37	48.10	35.12
Combined	17	0.0003	5	86.64	83.46	68.09	48.04	35.53
Combined	29	0.0003	5	86.65	83.37	67.19	47.82	35.51
Combined	43	0.0003	7	86.94	82.88	67.56	45.92	34.77
Random	17	0.001	3	86.04	80.96	65.48	44.84	29.33
Random	29	0.001	5	86.17	81.55	66.58	42.80	30.62
Random	43	0.001	3	86.58	82.36	65.73	42.64	29.90

Table 20: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

denominator, while its target is also zero.

Lemma 1 (Fixed-weight row gradient). *For each $j \in J$, $\partial\ell/\partial s_j = q_j - y_j$.*

Proof. Differentiating the finite sum gives $\partial D/\partial s_j = w_j e^{s_j}$. The logarithm contributes $w_j e^{s_j}/D = q_j$, and the linear target term contributes $-y_j$. $\square$

An individual positive derivative need not be negative. If $P = \{1, 2\}$ and $q = (0.8, 0.1, 0.1)$, the gradient is $(0.3, -0.4, 0.1)$. However,

$$\sum_{p \in P} \frac{\partial \ell}{\partial s_p} = - \sum_{n \in N} q_n \leq 0. \quad (11)$$

Thus claims about total positive attraction do not imply a sign for every positive score.

L.2 Expansion of the positive set

Proposition 1 (Target redistribution). *Assume unit weights on every candidate and fixed scores. Let P contain $K \geq 1$ positives. Relabel a disjoint set $A \subseteq J \setminus P$ of $M \geq 0$ already-present candidates as positive, yielding $P' = P \cup A$. For $p \in P$,*

$$\frac{\partial \ell'}{\partial s_p} - \frac{\partial \ell}{\partial s_p} = \frac{M}{K(K+M)}. \quad (12)$$

For $a \in A$, the difference is $-1/(K+M)$. For the remaining candidates, it is zero.

Proof. The scores, candidates, and weights are unchanged, so D and all q_j remain fixed. For an original positive, subtracting the two derivatives gives

$$(q_p - 1/(K+M)) - (q_p - 1/K) = \frac{M}{K(K+M)}. \quad (13)$$

A newly positive candidate changes target probability from zero to $1/(K+M)$. The remaining targets stay zero. The original positives jointly lose $M/(K+M)$ target probability, precisely the amount assigned to the new positives. $\square$

The premise that candidates are already present matters: adding new candidates changes the denominator as well. Likewise, changing weights changes q . The following identity separately accounts for a change in the eligible text anchors of symmetric training.

Equation 4 gives the corresponding reverse-anchor averaging identity and its assumptions.

L.3 A constant matching the normalizer

Proposition 2 (Matched negative mass). *Let N be nonempty under the common assumptions. Define*

$$c^* = \frac{\sum_{n \in N} w_n e^{s_n}}{\sum_{n \in N} e^{s_n}}. \quad (14)$$

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Smoothing	17	0.0001	1	82.04	83.61	68.59	55.26	38.99
Smoothing	29	0.0001	10	73.63	79.76	60.67	50.92	33.45
Smoothing	43	0.0001	1	82.03	83.48	68.64	55.36	38.81
Ranking	17	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	29	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	43	0.0001	0	79.47	81.21	68.95	56.32	39.35
Fixed	17	0.0003	6	79.74	82.97	66.09	50.34	31.88
Fixed	29	0.0003	0	79.47	81.21	68.95	56.32	39.35
Fixed	43	0.0003	0	79.47	81.21	68.95	56.32	39.35
Shuffled	17	0.0003	1	82.85	83.10	66.64	52.60	38.10
Shuffled	29	0.0003	1	83.97	83.81	66.89	52.14	38.14
Shuffled	43	0.0003	1	83.20	83.54	67.16	52.06	38.33
Median	17	0.0001	3	82.76	83.54	67.96	53.64	38.90
Median	29	0.0001	2	82.44	83.74	67.96	54.42	39.42
Median	43	0.0001	2	82.78	83.89	68.09	53.86	39.30
Constant	17	0.0001	1	82.00	83.88	68.59	55.36	39.48
Constant	29	0.0001	1	82.03	83.65	68.70	55.10	39.39
Constant	43	0.0001	2	82.43	83.48	67.82	55.16	39.31

Table 21: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

Comparator	Endpoint	Δ	99.17% interval	Seed 17	Seed 29	Seed 43
Expanded	e-SNLI-VE	-0.255	[-0.605, +0.065]	-0.109	-0.137	-0.520
Expanded	SugarCrepe	-0.155	[-0.509, +0.209]	+0.080	-0.173	-0.373
Neutral	e-SNLI-VE	+0.081	[-0.100, +0.273]	+0.221	+0.253	-0.230
Neutral	SugarCrepe	-0.155	[-0.371, +0.056]	-0.133	-0.120	-0.213
Contradiction	e-SNLI-VE	-0.535	[-0.890, -0.193]	-0.398	-0.484	-0.723
Contradiction	SugarCrepe	-0.093	[-0.442, +0.261]	+0.133	-0.120	-0.293

Table 22: Six primary contrasts: Combined minus the comparator, in percentage points. Intervals use paired image-cluster resampling and the fixed six-comparison Bonferroni family.

Compute c^* at the specified scores and hold it fixed for differentiation. Replace w_n by this constant for each $n \in N$, keeping positive weights one. At these scores, the new normalizer, loss, each positive probability and derivative, and total negative derivative equal their original values.

Proof. The denominator defining c^* is positive because N is nonempty and every exponential is positive. Therefore c^* exists, is nonnegative, and satisfies

$$\sum_{n \in N} c^* e^{s_n} = \sum_{n \in N} w_n e^{s_n}. \quad (15)$$

The positive exponential mass is unchanged. Adding it to both sides proves equality of the normalizers. The target term also stays fixed, giving equal losses. Each positive probability is e^{s_p}/D , so

its probability and derivative are equal. Finally, the total negative derivative is the negative exponential mass divided by D in each case. $\square$

Individual negative derivatives are generally unequal: they are $w_n e^{s_n}/D$ and $c^* e^{s_n}/D$. Matching preserves the total amount of competition, while changing its allocation. This is a row-specific analytical construction; a single global constant need not match every row. If instead $c^*(s)$ is recomputed differentiably, the new normalizer equals the original normalizer as a function of scores. Differentiating that equality restores the original individual negative derivatives. The allocation comparison therefore requires detachment.

Policy	Add A	Add O	Repl. A	Repl. O	Repl. R	Swap A	Swap O
Frozen	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Source	83.24	89.72	83.76	94.09	72.00	67.47	61.77
Expanded	87.86	90.74	82.91	93.02	72.00	65.12	60.68
Contradiction	88.05	90.59	83.04	92.88	72.02	64.96	60.27
Neutral	87.91	90.98	82.91	93.18	71.46	65.17	60.41
Combined	87.91	90.74	83.04	93.20	71.08	65.27	59.05
Random	86.85	89.19	81.09	92.39	68.66	63.06	57.14
Smoothing	80.54	87.20	83.46	93.48	72.43	67.62	62.99
Ranking	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Fixed	80.11	87.86	82.66	93.79	69.77	66.97	61.36
Shuffled	86.37	90.88	83.08	93.68	70.63	68.07	61.36
Median	85.12	91.21	83.71	94.05	71.03	68.22	62.31
Constant	84.20	90.80	83.46	94.23	72.00	67.57	62.31

Table 23: SugarCrep category means (percent). A, O, and R denote attribute, object, and relation. Category denominators are listed in Appendix E.

Seed	Queries	Pool	Lost	New	Net loss
17	I→T	5,000	649	253	396
17	T→I	25,000	2,195	1,132	1,063
29	I→T	5,000	832	293	539
29	T→I	25,000	2,651	1,241	1,410
43	I→T	5,000	597	248	349
43	T→I	25,000	1,976	1,045	931

Table 24: Retrieval success transitions from source positives to positive expansion. Lost and new count changed rank-one successes on the identical query pool.

The transpose need not preserve the match.

Row-specific matching does not generally preserve the normalizers of the transposed text-to-image objective. For example, take two images, three captions, all scores zero, positive pairs (1, 1) and (2, 2), and weights

$$W = \begin{pmatrix} 1 & 0.2 & 0.8 \\ 0.6 & 1 & 0.4 \end{pmatrix}. \quad (16)$$

Both row constants are 0.5 and both image-row losses remain $\log 2$. The two eligible text-column normalizers change from 1.6, 1.2 to 1.5, 1.5. Consequently the full symmetric loss increases by

$$\frac{1}{4} \log \frac{1.5^2}{1.6 \cdot 1.2} \approx 0.0396513. \quad (17)$$

A symmetric matched control therefore requires additional constraints. The executed fixed constant 0.25 is not a c^* -matched experiment.

L.4 Suppression does not control every negative group

For one positive, let all negative weights lie in $[0, 1]$. Write

$$S = \sum_j e^{s_j}, \quad D = e^{s_+} + \sum_{n \in N} w_n e^{s_n}. \quad (18)$$

The inequality $D \leq S$ implies $q_+ = e^{s_+}/D \geq p_+ = e^{s_+}/S$. Hence total negative derivative satisfies

$$\sum_{n \in N} q_n = 1 - q_+ \leq 1 - p_+. \quad (19)$$

This bound does not hold for every subset of negatives. Consider exponential masses 1 for the positive, 1 for a chosen negative group, and 100 for other negatives. Weight the chosen group by 0.5 and the others by 0.001. The chosen group’s probability changes from

$$\frac{1}{102} \approx 0.009804 \quad \text{to} \quad \frac{0.5}{1 + 0.5 + 0.1} = 0.3125. \quad (20)$$

Policy	I→T R1	R5	R10	T→I R1	R5	R10
Frozen	56.32	79.82	87.12	39.35	65.40	75.62
Source	55.85	79.59	86.93	39.38	65.67	75.91
Expanded	47.29	71.56	80.89	34.84	60.77	71.50
Contradiction	47.25	71.61	80.87	34.81	60.78	71.46
Neutral	48.01	72.58	81.79	35.39	61.56	72.19
Combined	47.26	71.93	81.33	35.27	61.36	72.05
Random	43.43	67.84	77.81	29.95	55.28	66.61
Smoothing	53.85	77.94	85.71	37.08	63.47	73.72
Ranking	56.32	79.82	87.12	39.35	65.40	75.62
Fixed	54.33	77.83	85.75	36.86	62.66	73.19
Shuffled	52.27	76.63	84.83	38.19	64.41	74.58
Median	53.97	77.93	85.83	39.21	65.42	75.66
Constant	55.21	78.85	86.55	39.39	65.62	75.85

Table 25: Full-pool COCO recall at ranks one, five, and ten, averaged across selected seeds (percent).

Input	Probabilities	Acc.	Macro-F1	ECE	Brier	NLL
Image	uncalibrated	35.64	34.06	4.73	0.674	1.111
Image	calibrated	35.64	34.06	0.05	0.666	1.098
Text	uncalibrated	61.76	61.38	5.78	0.509	0.879
Text	calibrated	61.76	61.38	0.68	0.503	0.858
Multimodal	uncalibrated	65.69	64.96	12.53	0.487	0.887
Multimodal	calibrated	65.69	64.96	1.38	0.457	0.783

Table 26: Held-out relation classification on the newly sampled test set. Accuracy, macro-F1, and ECE are percentages; Brier and NLL retain their native scale.

The total negative mass decreases, but that group’s mass increases. Contradiction weighting by two also lies outside the premise $w_n \leq 1$.

L.5 Summed-positive probability

Let $A = \sum_{p \in P} e^{s_p}$. A different objective is

$$\ell_{\text{set}} = \log D - \log A. \quad (21)$$

For $p \in P$, direct differentiation gives

$$\frac{\partial \ell_{\text{set}}}{\partial s_p} = q_p - \frac{e^{s_p}}{A} \leq 0, \quad (22)$$

because $D \geq A$. The attraction is allocated according to the positive scores, rather than equally across positives. This permits the highest-scoring positives to dominate the target distribution. The identity describes a distinct objective, not an equivalent implementation of Equation 10.

L.6 Parameter-dependent weights and score updates

Let J_+ be a fixed nonzero-support set containing P . For $j \in J_+$, assume that $w_j(\theta) > 0$ is locally

differentiable, with $w_p(\theta) \equiv 1$ on positives. All other weights are fixed at zero. With fixed targets and differentiable parameter-dependent scores,

$$\begin{aligned} \nabla_{\theta} \ell &= \sum_{j \in J_+} (q_j - y_j) \nabla_{\theta} s_j \\ &+ \sum_{j \in J_+} q_j \nabla_{\theta} \log w_j. \end{aligned} \quad (23)$$

The second term vanishes for detached weights. No logarithmic derivative is taken at a zero weight; the identity does not describe a change of support through zero. Even with detached weights, a parameter update combines the score Jacobians across every candidate and batch. A sign or magnitude bound for an individual score derivative is not a corresponding bound on the parameter-gradient norm or on the score after optimization.

L.7 Other fixed-score identities

The following facts collect the single-positive special cases behind similarity suppression. Let $0 \leq$

Input	Gate	Accept	Correct	Coverage	Precision	95% interval
Image	supported	0	0	0.00	n/a	n/a
Image	contradicted	0	0	0.00	n/a	n/a
Text	supported	0	0	0.00	n/a	n/a
Text	contradicted	0	0	0.00	n/a	n/a
Multimodal	supported	0	0	0.00	n/a	n/a
Multimodal	contradicted	683	647	11.65	94.73	[92.72, 96.59]

Table 27: Selective test predictions. Precision is undefined when no pair is accepted. Intervals resample whole test images and are descriptive.

Input	C	Temp.	Max. iter.	OOF F1
Image	0.1	3.085	1182	33.17
Text	1	1.312	2318	63.46
Multimodal	0.01	1.904	2505	67.22

Table 28: Classifier selection and completion. Maximum iteration count is over the full-training grid; OOF is image-disjoint out-of-fold macro-F1 (percent).

$w_n \leq 1$ for negatives, $w_+ = 1$, and

$$R = \sum_{n \in N} (1 - w_n) e^{s_n}, \quad D = S - R. \quad (24)$$

Since $0 \leq R < S$,

$$\log q_+ - \log p_+ = -\log(1 - R/S) \geq R/S \geq 0. \quad (25)$$

The logarithm inequality follows by integrating $1/(1-x) \geq 1$ over $[0, R/S]$. It compares probabilities at fixed scores and does not change their raw ranking. For a group of m negatives at common score a and common weight w , its gradient mass is

$$G = \frac{mwe^a}{D}. \quad (26)$$

The denominator also depends on the group. Let A be the unweighted exponential mass outside that group, so $S = A + me^a$. Nonnegative suppression elsewhere gives

$$R \geq m(1-w)e^a, \quad \frac{R}{S} \geq \frac{m(1-w)e^a}{A + me^a}. \quad (27)$$

This rational lower bound tends to $1-w$ as m grows with A, a, w fixed. The actual log-probability gain need not saturate; with $w = 0$ and no other suppressed candidates, it can grow without bound as m increases. At equal scores, a smaller weight gives smaller normalized probability within the same denominator. This says nothing about whether the weighting identifies a correct semantic relation.

As every weight tends to one, the weighted loss converges to the corresponding unweighted uniform-positive loss by continuity. A common negative weight below one does not yield the ordinary loss while the positive weight remains one. Finally, weighted normalization is ordinary softmax normalization on shifted scores $s_j + \log w_j$, with $\log 0 = -\infty$. It is not label smoothing, which changes the targets.

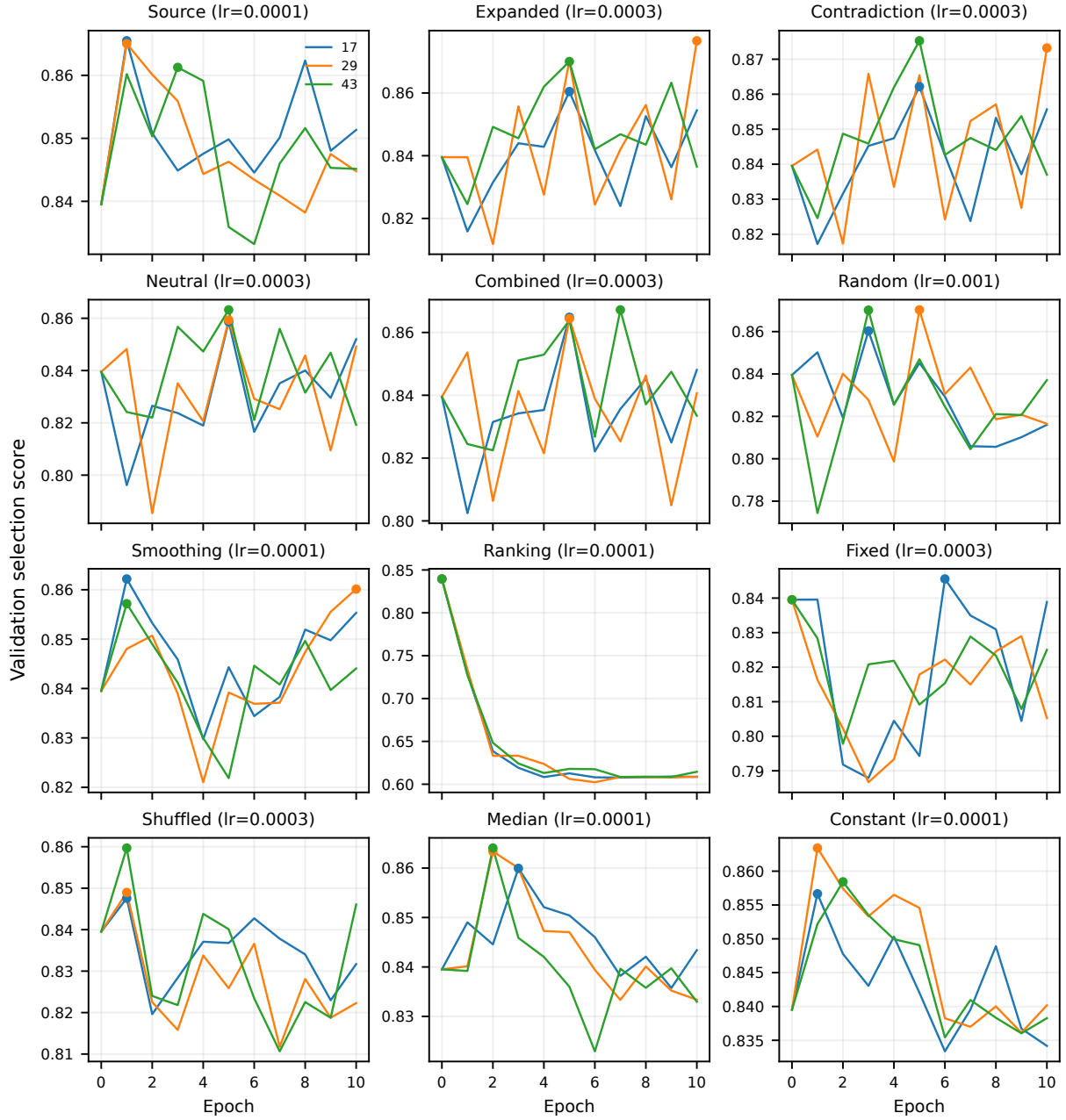


Figure 5: Validation trajectories at the selected learning rate of each policy. Lines are seeds; dots mark retained epochs. Every candidate evaluated epoch zero through ten. These are selection data, not held-out outcomes.